Q 3 (i)(a)

A 3 (i)(a)

In biology, domain means largest of all groups in the ~~b~~ classification of life. Domain is the group of all kingdoms or ~~above all~~ taxanomic category above all kingdoms. In 1990, carl woese introduced a three domain system of classification. The three domain system are :-

~~Arch~~ Archea.
Bacteria
Eukarya.

Classification into 3 domain is based on the difference of the sequence of nucleotides in the rRNA (ribosomal ribonucleic acid) of the cell, cell membrane lipid structure and it's sensitivity to antiboitics.

* Domain Archea :-

Following are the characteristics of domain archea :-

Domain Archea are prokaryotic cell. Their cell wall contains no peptidoglycan. Their rRNA (ribosomal RNA) is not found in Bacteria and Eukarya.

(4) Archea are not sensitive to some antiboitics that affect the bacteria, they are ~~the~~ sensitive to some antiboitics that affect the eukarya.

(5) Archea often live in extreme environment

(6) Archea membrane can withstand higher temperature and stronger acid concentrat

(7) Archeal creatures that include :-

(1) Methanogens
(2) Halophiles
(3) Thermoacidophiles.

★ Domain Bacteria :-
Following are the characteristics of domain Bacteria are :-

1. Domain Bacteria are prokaryotic cells.
2. Their cell wall contains peptidoglycan.
3. ~~Their rRNA~~ They have rRNA that is uni to Bacteria.
4. Domain Bacteria are sensitive to some tranditional antibacterial antiboities but are resistant to some antiboitics that affect the eukarya.

* Domain Eukarya:-

Following are the characteristics of domain Eukarya are:-

① Domain Eukarya have eukaryotic cells.

② Not all eukarya have cell wall, their cell wall does not contains peptidoglycan.

③ Their rRNA is unique to eukarya.

④ Eukarya are resistant to some traditional antibacterial antiboitics, these are sensitive to some antiboitics that affect the eukaryotic cells.

Domain Eukarya

Animalia Plantae Fungi Protista

Domain Eukarya is further divided into 4 Kingdoms

Animalia:-

It is a multicellular heterotrophic organisms. which can generally move from place to place. They can hunt their own food They are called producers There are vertebrates ÷ insects, starfish etc. and invertebrates :- humans, frogs etc.

Plantae :-

It is multicellular autotrophic organisms which will create their own food. for example mango tree etc.

Fungi :-

It is multicellular mostly organism which are heterotrophic. like mushroom.

Protista :-

It is unicellular eukaryotic autotrophic and heterotrophic organism

Q.3(ii)
A.3(ii)

The nucleus is the control centre of the cell, containing the genetic information (DNA) and directly all cellular activities.

① Nuclear envelope:-
Double membrane surrounding the nucleus. outer membrane is continous with rough endoplasmic reticulum. controls entry / exit of materials.

② Nuclear pores:-
Small, openings in the nuclear envelope that allows mRNA, ribosomes and other molecules to move b/w nuclear and cytoplasm.

③ Nucleoplasm:-
fluid matrix inside the nucleus in which chromatins and nucleolus is suspended.

④ Nucleolus:-
Dense body inside nucleus, site of rRNA synthesis which combines with protein to form ribosomes.

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Q.3(iii)(or)

A:3(iii)(or)

Carbohydrates are classified on the basis of the numbers of sacchride (monosacchride) units in the molecule. General formula :-

$C_n(H_2O)_y$; ratio of C : H : O :-
1 : 2 : 1

It's classified into three types :-

* Monosaccharide :-
It is the simplest carbohydrates, it is a single sacthride unit ; cannot be hydrolysed further, Sweet and crystalline and it is soluble in water. Example:- Glucose, fructose, galactose, ribose etc.
Molecular formula :-
Glucose :-
$C_6H_{12}O_6$

Ribose :-
$C_5H_{10}O_5$

* Disacchrides :-
It is form by two monosacchrides bonded by condensation (releasing water) it is slightly less sweet.
Molecular formula :-
$C_{12}H_{23}O_{11}$

Sucrose :- cane sugar
Glucose + Fructose $\longrightarrow C_{12}H_{22}O_{11} + H_2O$

Maltose :- Malt sugar
Glucose + Glucose $\rightarrow C_{12}H_{22}O_{11} + H_2O$

Lactose :- (milk sugar)
Galacose + glucose $\rightarrow C_{12}H_{22}O_{11} + H_2O$

Q:3(iv)(08)
A:3(iv)(08)

Effect of temperature :-
As temperature increases from Low to optimum, enzyme active increases because molecules move faster, optimum temperature ~~increasing collision frequency b/w enzyme~~ :- 36°c — 38°c (body temperature)
Enzyme works fastest here

Above optimum :-
Atoms of enzymes vibrate vigourously, globular structure and active site are damaged. This is called denaturation. Activity falls sharply to 0.

At very low temperature :-
Enzymes is inactive due to insufficient activation energy

Activity resumes when normal temperature is restored.

Effect of Substrate concentration:-

When enzymes molecule are in ~~excess~~ excess, increasing substrate concentration increases reaction rate because more substrate molecules fill ~~is~~ the available active site.

* Up to a certain point, rate is directly proportional to substrate concentration

* Beyond a maximum :- All active sites are occupied called saturation of active sites. Further addition of substrate has no effect on rate.

Section - B

Q:2 (i) (or)

A:2 (i) (or)

Medicine / surgery :-
- It's degree is MBBS, it diagnoses and treats ~~human~~ human disease and disorders

Biotechnology :- Studies organisms to develope useful products (vaccines, medi... etc)

Agriculture / farming :- Applies biological knowledge to improve crop yeild fertility for food.

Q:2 (ii) (or)

A:2 (ii) (or)

Ronald Ross :- In 1897, A british physician who fed female Anopheles mosquitoes on malaria-infected patients, then dissected the mosquitoes and found plasmodium in their stomach walls. His hypothesis :- the Anopheles mosquitoes is the vector transmit the parasite

Q:2(iii)(or)
A:2(iii)(or)

Domain Archea :-
Following are the characteristic of domain Archea :-
i) Domain Archea are prokaryotic cells.
ii) Their cell wall doesnt contain peptidoglycan.
iii) Their rRNA does not found in Bacteria and Eukarya.
iv) They often live in extreme environment.

In 1990, Carl woesse introduced a domain system which ~~introduced~~ include:
① Archea ② Bacteria ③ Eukarya.

Q:2 (iv)(or)
A:2 (iv)(or)

~~Chloro~~ Chloroplast	Mitochondria
It has double membrane, inners membrane form thylakoids grouped into grana, fluid filled stroma outside	It has double membrane, inner membrane form cristae, fluid filled matrix inside.
It's function is photosynthesis, convert light energy into chemical energy	ATP production, release energy from glucose

It is found in plant cell only	It is found in eukaryotic cells

Q2(v)
A:2(v)

Metaphase :-
Chromosomes move to the ~~equta~~ equatorial plate (middle of cell, Spindle fibres from both Poles attach to the centromere of each chromosome, aligning them precisely. This is the stage where chromosomes are more visible.

Anaphase :-
The centromere spilt, separating each chromosome into two chromatids. Spindle fibres shorten and pull the chromatids to opposite poles of the cell. By the end of the anaphase each pole has a complete set of chromosomes.

Q:2(vi)

A.2(vi)

Prophase I is the longest phase of meoisis:

1) Chromatin condenses into chromosomes and the nuclear membrane begins to break down.

2) Synpsis :- homologous chromosome come together and pair up along their entire length

3) Each pair of homologous chromosomes form a bivalent (tetrad of 4 chromatids :- 2 Chromosomes X 2 chromatides each)

4) Crossing over :- non-sister Chromatids of homologous Chromosomes exchange segments at points called chaismata, creating new gene combinations and contributing to genetic variation.

Q:2(vii)

A:2(vii)

Homoeostatsis :-

is the tendency of living organisms to maintain a stable, relatively constant internal environment.

Q.2(viii)
A:2(viii)

Shoot system:
Located above the ground. Consists of stem, leaves and reproductive organs (flowers, fruits, seeds). Functions include photosynthesis (leaves). transport of food and water through xylem and phloem and reproduction.

Root system:
Located below grond. Consist of primary and secondary roots.

Q:2(ix)(or)
A:2(ix)(or)

1. Energy storage:
Lipids store. Long term energy in fat cell and liver.

2. Cell membrane component:
phospholipids and cholest. form the plasma membrane bilayer.

4. Hormone synthesis:
Steroid lipids form hormones ~~and~~ like testoterone and aldoterone.

Insulation :-
lipid acts as insultators against atmospheric heat and cold.

Q: 2(x)(or)
A: 2(x)(or)

A nucleotide is the monomer of nucleic acids (DNA and RNA). It has 3 components:-

① Pentose suger
② Nitrogenous base.
③ Phosphate group.

Q: 2(xi)(or)
A: 2(xi)(or)

Many enzyme require a non-protein helper a cofactor. Three types:-

① Activator:-
Detachable inorganic ions that activate the enzymes

② ~~Pro~~ Prosthetic group:-
An organic molecule ~~perma~~ ~~nent~~ Permanently and tightly attached to the enzyme.

Coenzyme:- A detachable organic cofactor that temporarily binds the enzyme.

MCQs:-

1. C
2. C
3. B
4. C
5. B
6. C
7. B
8. B
9. C
10. C
11. B
12. C